



SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1047 - Advanced Algorithms & A.I. Search

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,3

EXAMINER(S):

Dr.Mark Humphrys	(Internal)	(Ext:8059)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: See inside.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

CSC1047 - Advanced Algorithms and AI Search - 2024-25

Multiple choice

Answer all 20 multiple-choice questions. (2 marks each.) Total 40 marks.

No negative marking.

Only the filled-in boxes will be looked at. No additional commentary or analysis will be read.

Remember to hand up the multiple-choice answer sheet!

1. We have a neural network where inputs are finite (positive or negative). If a node has a (negative) infinite threshold, then:
 1. If weights are positive, you have constant output 0.
 2. If weights are finite, you have constant output 1.
 3. If weights are finite, you have constant output 0.
 4. None of the above.
 5. If weights are positive, you have constant output 1.
 6. If weights are negative, you have constant output 1.
 7. If weights are negative, you have constant output 0.

2. An upper bound on the *game-tree complexity* of Xs and Os is:
 1. 9^3
 2. $3!$
 3. 9^9
 4. 3^3
 5. $9!$
 6. 3^9

3. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. We
$$\frac{\delta E}{\delta w}$$
consider E, the error in our solution, and $\frac{\delta E}{\delta w}$ the direction in which to change our solution. Which of the following is the case:

1. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, neither E nor $\frac{\delta E}{\delta w}$ are known.
2. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known but $\frac{\delta E}{\delta w}$ is unknown.
3. For Neural Networks, E is known but $\frac{\delta E}{\delta w}$ is unknown. For GAs, neither E nor $\frac{\delta E}{\delta w}$ are known.
4. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known and $\frac{\delta E}{\delta w}$ is known.

4. We are learning from exemplars. We experience 3 exemplars:
 When input is (x1) output should be (a1).
 When input is (x2) output should be (a1).
 When input is (x1) output should be (a2).
 How can this happen?

1. The 2nd exemplar can happen but not the 3rd.
2. The 3rd exemplar can happen but not the 2nd.
3. There are missing inputs, or the world is probabilistic.
4. This cannot happen.
5. The machine is badly programmed.
6. This always happens.

5. In machine evolution, we have 4 individuals, where individual i has fitness $f(i)$:

i	1	2	3	4
$f(i)$	0	5	4	5

If we use a "Soft max" probability distribution with a "temperature" parameter T to choose who reproduces, then as $T \rightarrow \text{infinity}$ the probability of each individual reproducing goes to:

1.	0	1/2	0	1/2
2.	1/4	1/4	1/4	1/4
3.	0	1	0	0
4.	0	1	0	1
5.	0	0.9	0.1	0.9
6.	0	5/14	4/14	5/14

6. The input to the 7th output node of a neural network is:

$$x_7 = w_{17} y_1 + w_{27} y_2 + w_{37} y_3$$

Then $\partial x_7 / \partial w_{27}$ is:

1. y_j

$$\sum$$

2. $j \ w_{jk}$

3. w_{27}

4. y_2

$$\sum$$

5. $j \ y_j$

6. w_{jk}

7. In a perceptron, let inputs be 0 or 1. Let the output be $y = 0$ or 1 , and the correct output be C . The Perceptron Learning Rule can be summarised as:

1. If $y=0$ and $C=1$, then reduce threshold and reduce weights.
2. If $y=0$ and $C=1$, then reduce threshold and increase weights.
3. If $y=1$ and $C=0$, then reduce threshold and increase weights.
4. If $y=0$ and $C=1$, then increase threshold and reduce weights.
5. If $y=1$ and $C=0$, then increase threshold and increase weights.

8. Given a line in 2-dimensional space:

$$y = a x + b$$

Define a perceptron that receives x and y as input and fires if and only if the point is on this side of the line:

$$y \leq a x + b$$

1. weights: $-a, 1$, threshold: b
2. weights: $a, -1$, threshold: $-b$
3. weights: $a, -1$, threshold: b
4. weights: $-a, 1$, threshold: $-b$

9. We do machine evolution as follows:

Start with 100 random solutions.

Repeat:

Test all 100 for fitness.

Pick the 10 best.

Make 10 copies of each to construct the next generation.

Loop again.

The problem with this is:

1. We should pick according to probabilities rather than an absolute number.
2. We lose the best solutions after finding them.
3. It will end up focused on one solution.
4. 10 is too small.
5. There is no problem.
6. 100 is too small.
7. Too many solutions reproduce.
8. No new solution is ever made.

10. In a neural network, the network generates a (vector) output $y = (y_1, \dots, y_n)$. We then tell it the "correct" output $O = (O_1, \dots, O_n)$. Consider the following as a measure of "error" E :

$$E = \sum_k (y_k - O_k)$$

An example showing that this is this a bad measure of the error is:

1. $y = (5, 0, -5)$, $O = (-5, 0, -5)$
2. $y = (-5, 0, 5)$, $O = (-5, 0, 5)$
3. $y = (5, 0, -5)$, $O = (-5, 0, 5)$
4. $y = (5, 0, 5)$, $O = (-5, 0, 5)$
5. $y = (5, 0, 5)$, $O = (-5, 0, -5)$
6. $y = (5, 0, 5)$, $O = (5, 0, 5)$

11. We are learning to represent a function $Q(x)$. The input x is a 10 dimensional vector. Each dimension is an integer from 1 to 100. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:
1. 100
 2. 1000
 3. 10^{100}
 4. 100^{10}
 5. 110
 6. 10
 7. 1010
12. With large state-spaces, it is hard to draw the entire state-space representation of a problem. This is solved by:
1. We draw a partial state-space representation of the problem.
 2. It is not solved. We do not draw the state-space representation of the problem.
 3. Writing a program to automatically draw it.
 4. Using symmetry to reduce the size of the state-space.
 5. Using a heuristic to reduce the size of the state-space.
 6. Doing a truncated search.
13. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. Which of the following is the case:
1. Neural Networks make a random move, GAs make a random move.
 2. Neural Networks make a random move, GAs make a directed move.
 3. Neural Networks make a directed move, GAs make a directed move.
 4. Neural Networks make a directed move, GAs make a random move.

14. We are learning to represent a function $Q(x)$. The input x is a 100 dimensional vector. Each dimension is an integer from 1 to 10. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:
1. 10
 2. 10^{100}
 3. 100^{10}
 4. 110
 5. 1000
 6. 1010
 7. 100
15. In traversing a search tree, nodes that are placed on the tree to the left are searched before nodes placed to the right. Does the ordering of the nodes introduce a bias?
1. Yes. But it is never a problem.
 2. No.
 3. Best-first search will eliminate all bias.
 4. Yes. But a heuristic can be used to produce a bias that is better than random.
 5. Yes. But it is rarely a problem.
 6. Yes. But a heuristic can be used to eliminate all bias.
 7. Not if done right.
 8. Yes. But it is alright because all nodes will be searched eventually.
16. Consider the function $y = f(x) = \mathcal{S}(x)$ where $\mathcal{S}$ is the sigmoid function. f has its (a) highest slope value, and (b) steepest slope (highest absolute value) at:
1. (a) $y = 0$ or 1 , (b) $y = 0$ or 1
 2. (a) $y = 0$ or 1 , (b) $y = 1/2$
 3. (a) $y = 1/2$, (b) $y = 1/2$
 4. (a) $y = 1/2$, (b) $y = 0$ or 1

17. In neural networks, we need thresholds. Otherwise if all inputs are 0, then output of every single hidden node is:

1. infinity
2. 1
3. -infinity
4. $1/2$
5. 0

18. One of these is *known* to be untrue:

1. AI is possible.
2. In the future, immortal humans will travel to distant stars.
3. Computers will never be better than humans at football.
4. Computers will never be better than humans at chess.
5. AI (the construction of intelligent machines) is impossible.
6. In the future, intelligent machines will exterminate the human race.
7. In the future, humans will live forever on earth in artificial bodies, regularly backing up their brains in order to cheat death.

19. In machine evolution, each individual has a genotype encoded in binary format. We repeatedly select 2 parents to make 2 children by copying the 2 genotypes, with the possibility of mutation and crossover. Let p_c be the probability of crossover between the 2 genotypes. Let p_m be the probability that any bit in the genotype mutates when it is inherited.

Each generation, we only pick the *single* best individual to be *both* parents of each child. Then:

1. This is a normal search.
2. We rely on mutation to produce new solutions.
3. No new solutions are produced.
4. Both mutation and crossover will produce new solutions.
5. We rely on crossover to produce new solutions.

20. In neural networks, let $\frac{\delta E}{\delta w}$ be the change in error as we change a weight. Let c be a positive constant. Then a rule to reduce error would be:

1. $W := -c \frac{\delta E}{\delta w}$
2. $W := W + c \frac{\delta E}{\delta w}$
3. $W := W - c \frac{\delta E}{\delta w}$
4. $W := c \frac{\delta E}{\delta w}$

Long questions

Answer 2 long questions out of 4. (25 marks each.) Total 50 marks.

- Each question gives **maximum number of sentences** for the answer. Do not exceed this.
- You might like to write your answers as **rough work** first to figure out what sentences to use. Make sure any rough work is crossed out.
- Put your sentences as **bullet points** not a paragraph.
- If you answer more than 2 questions, I will pick the best 2.

Question 1

[Total marks: 25]

- 1(a) Is this function $y = f(x)$ easy or hard to maximise from *exemplars*? [5 marks]
 $f(1.0372) = 80$.
 $f(1.0373) = 100$.
 $f(x) = 0$ for all other x .
 Maximum length: **3 sentences**
- 1(b) Consider a heuristic search as follows: [5 marks]
 Initialise with 20 random solutions.
 Repeat for many iterations:
 {
 Test the 20 solutions.
 Pick the best n .
 Make small changes in them
 to generate 20 new solutions.
 }
 If $n=19$ what will this look like?
 Maximum length: **2 sentences**

- 1(c) In normal software engineering, the engineer seeks to find the correct solution and code it as an algorithm. [5 marks]
 In AI, the engineer may deliver a program that fails badly on occasion, and is pretty much known *not* to be the best solution.
 How can this be justified?
 Maximum length: **2 sentences**
- 1(d) Is this function $y = f(x)$ easy or hard to maximise from *exemplars*? [5 marks]
 $f(a) = 1$ for some value a .
 $f(x) = 0$ for all other x .
 Maximum length: **3 sentences**
- 1(e) Is this function $y = f(x)$ easy or hard to maximise from *exemplars*? [5 marks]
 $f(a) = 0$ for some value a .
 $f(x) = 1$ for all other x .
 Maximum length: **3 sentences**

Question 2

[Total marks: 25]

- 2(a) Can informed (heuristic) search be worse than uninformed (random) search? [5 marks]
 Maximum length: **1 sentence**
- 2(b) In the N-puzzle problem, we use a heuristic (distance to goal estimate) as follows: $h(n)$ = number of pieces out of place. [5 marks]
 If $h(x) = 1$ then are we close to the goal?
 Maximum length: **4 sentences**
- 2(c) In a very small search tree, it may not matter what search algorithm we use. Why? [5 marks]
 Maximum length: **1 sentence**
- 2(d) Imagine the following situation in Best-first search. We go down a path where the heuristic evaluation function tells us we are 2 steps away from the goal - and no matter how far down this path we go, the heuristic evaluation function still tells us we are 2 steps away from the goal. Can this happen and is it a problem? [5 marks]
 Maximum length: **3 sentences**
- 2(e) The idea of "informed search" is that we choose the next state that is *most likely* to lead to the goal. But if we can do that, then the problem is solved! Just keep choosing the best next state after that and you get to the goal ASAP. No search needed! No AI needed. Explain. [5 marks]
 Maximum length: **2 sentences**

Question 3

[Total marks: 25]

- 3(a) Given a line in 2-dimensional space: [5 marks]

$$a x + b y = c$$

Define a perceptron that receives x and y as input and fires if and only if the point is on this side of the line:

$$a x + b y \geq c$$

Maximum length: **2 sentences**

- 3(b) Given a line in 2-dimensional space: [5 marks]

$$a x + b y = c$$

Define a perceptron that fires if and only if the point is on this side of the line:

$$a x + b y \leq c$$

Maximum length: **3 sentences**

- 3(c) A mobile robot has Input state = (angle of direction of nearest wall, [5 marks]

velocity value). Output behaviour = one of (turn left, turn right, stop, keep going).

The Input state is clearly missing some information. What are the consequences of this missing information?

Maximum length: **3 sentences**

- 3(d) Consider the character recognition problem for handwriting. [5 marks]

If a character is isolated from other characters, there may be a problem with *missing inputs* when we try to recognise it. Explain.

Maximum length: **3 sentences**

- 3(e) A multi-layer neural network has 1 input unit, n hidden units, and 1 [5 marks]

output unit. The weights on the input layer are all the same: $w_{ij} = W_1$.

The thresholds of the hidden units are all the same: $t_j = T$. The weights on the output layer are all the same: $w_{jk} = W_2$.

Show that this network is equivalent to a network with 1 input unit, 1 hidden unit, and 1 output unit. What are the weights and thresholds of the latter network?

Maximum length: **5 sentences**

Question 4

[Total marks: 25]

- 4(a) In a genetic algorithm (GA), let the probability of an individual being [5 marks]

picked for reproduction be:

$$p = \frac{\text{my fitness}}{\text{sum of fitnesses of all creatures}}$$

Fitness cannot be zero. Why?

Maximum length: **1 sentence**

- 4(b) In a genetic algorithm (GA), let the probability of an individual being [5 marks]

picked for reproduction be:

$$p = \frac{\text{my fitness}}{\text{sum of fitnesses of all creatures}}$$

Fitness cannot be both positive and negative. Why?

Maximum length: **1 sentence**

- 4(c) In a genetic algorithm: What would happen if you had a *variable* (rather than a constant) population size? [5 marks]

Maximum length: **3 sentences**

- 4(d) In a genetic algorithm (GA), let each individual i have fitness $f(i)$, and let its probability of reproducing, $p(i)$, be: [5 marks]

$$p(i) = \frac{e^{\frac{f(i)}{T}}}{\sum_j e^{\frac{f(j)}{T}}}$$

where T is the "temperature".

Consider the individuals:

i	1	2	3	4
$f(i)$	0.2	1	0.2	1

What happens as $T \rightarrow 0$?

Maximum length: **4 sentences**

- 4(e) In a genetic algorithm: What would happen if rather than picking the fittest, you define a *minimum* fitness that individuals have to reach in order to reproduce? [5 marks]

Maximum length: **2 sentences**

Bonus question

Bonus question. (10 marks.)

It is "bonus" because I tell you it in advance. The question is to make you do some reading. You can prepare this question in advance.

Bonus question:

Tell me about some AI-related algorithm or software that was *not* mentioned on the course.

CSC1047 Answer sheet

Black out the square that matches the correct answer.

Exam No: _____

Class: _____

Seat No: _____

Question	1.	2.	3.	4.	5.	6.	7.	8.
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